GEON CHOI

Ph.D. student @ KAIST AI specializing in multimodal AI and AI for healthcare

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github.com/checkoneee
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RESEARCH INTERESTS

Mulimodal AI: Multimodal Large Language Models (MLLMs) and Vision-Language Action (VLA) Models
AI for Healthcare: Medical VLMs, Medical Imaging

PUBLICATIONS

C: Conference, **J**: Journal, **P**: Preprint, *: Equal Contribution

CHIL 2026 **[C.3] Language: A Benchmark for Structured and Sequential Chest X-ray Interpretation**
Jong Hak Moon, Geon Choi, Paloma Rabaey, Min Gwan Kim, Hyuk Gi Hong, Jung-Oh Lee, Hangyul Yoon, Eun Woo Doe, Jiyoun Kim, Harshita Sharma, Daniel C. Castro, Javier Alvarez-Valle, Edward Choi
Conference on Health, Inference, and Learning, 2026
Links: [arXiv] [Code]

CVPR 2026 **[C.2] Instruction-Guided Lesion Segmentation for Chest X-rays with Automatically Generated Large-Scale Dataset**
Geon Choi*, Hangyul Yoon*, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Eunho Yang, Edward Choi
IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026
Links: [arXiv] [Code]

NeurIPS 2025 **Spotlight** **[C.1] CXReasonBench: A Benchmark for Evaluating Structured Diagnostic Reasoning in Chest X-rays**
Hyungyung Lee, Geon Choi, Jung-Oh Lee, Hangyul Yoon, Hyuk Gi Hong, Edward Choi
Conference on Neural Information Processing Systems (Datasets and Benchmarks Track), 2025
Links: [arXiv] [Code]

Preprint **[P.2] XFlow: A Workflow Model for Instruction-Guided Lesion Segmentation in Chest X-rays**
Geon Choi, Hangyul Yoon, Hyunki Park, Sang Hoon Seo, Edward Choi
Arxiv, 2026
Links: [arXiv]

Preprint **[P.1] CheXpercept: A Benchmark for Evaluating Expert-Level Lesion Perception in Chest X-rays**
Geon Choi, Hangyul Yoon, Nalee Kim, Jeong Yun Jang, Hyunju Shin, Hyunki Park, Sang Hoon Seo, Edward Choi
Arxiv, 2026
Links: [arXiv]

EDUCATION

2/2026 - Present **Korea Advanced Institute of Science and Technology (KAIST)** **Seongnam, South Korea**
Ph.D., Artificial Intelligence (Advisor: Prof. Edward Choi)

2/2024 - 2/2026 **Korea Advanced Institute of Science and Technology (KAIST)** **Seongnam, South Korea**
M.S., Artificial Intelligence (Advisor: Prof. Edward Choi)

2/2020 - 2/2024 **Pohang University of Science and Technology (POSTECH)** **Pohang, South Korea**
B.S., Electrical Engineering
GPA: 4.04/4.3 (Summa Cum Laude)
Ranked 2nd in the department of electrical engineering.

3/2018 - 2/2020 **Jeonbuk Science High School** **Iksan, South Korea**
Early Graduation
Ranked 3rd, earning the POSTECH Presidential Fellowship.

EXPERIENCE

6/2023 – 8/2023 **KAIST AI Research Internship (KAIRI)** **Daejeon, South Korea**
- BioImaging, Signal Processing and Learning Lab, KAIST (Advisor: Prof. Jongchul Ye)
- Research Topic: Generative Models for Solving Inverse Problems in Medical Imaging

1/2023 – 6/2023 **Undergraduate Research Internship** **Pohang, South Korea**
- Bio Optics and Acoustics Lab, POSTECH (Advisor: Prof. Chulhong Kim)
- Research Topic: Computer-Aided Diagnosis System for Breast Ultrasonography

HONORS AND AWARDS

2024 - Present **KAIST Scholarship** — **KAIST**
Tuition Scholarship

2023 **KEC Scholarship** — **KEC Foundation**
Merit Scholarship for Outstanding EE Student ($2k)

2022 - 2023 **National Science & Technology Scholarship** — **KOSAF**
Full Tuition Scholarship

2020 - 2021 **POSTECH Jigok Scholarship** — **POSTECH**
Full Tuition Scholarship

2020 - 2023 **POSTECH Presidential Fellowship** — **POSTECH**
Merit Fellowship for Exceptional Freshman ($10k)

TEACHING

2022 **Course Assistant for Digital System Design (EECE273)** — **POSTECH**
Student Mentor

SERVICES

2024 - 2025 **KoSAIM Summer School Hands-on Session** — **KoSAIM**
Instructor: taught chest X-ray classification with CNN for medical AI researchers and medical students in South Korea.

LANGUAGES

English - professional working proficiency
Korean - native

SKILLS

Programming Languages: Python, C, MATLAB
Core Frameworks & Libraries: PyTorch, Transformers, scikit-learn
MLLM Training & Serving: TRL, verl, vLLM, DeepSpeed
Embodied AI: starVLA
Tools & Infrastructure: Git, Linux, TensorBoard, Wandb

REFERENCES

2/2024 - Present **Edward Choi, Associate Professor of KAIST** — **edwardchoi@kaist.ac.kr**
M.S. and Ph.D. Advisor